Apply filters to SQL queries

Project description

In this project, I investigated a potential security incident by analyzing organizational database records for suspicious behavior, Using MariaDB SQL syntax, I applied advanced filtering techniques including `AND`, `OR`, `NOT`, and `LIKE` operators to isolate specific subsets of data from the login logs and employee records. This analysis allowed me to identify vulnerable machines, track down unauthorized after-hours login attempts, and isolate geographic network patterns to enhance overall system security.

Retrieve after hours failed login attempts

To investigate a potential insider threat or external brute-force attempt after standard operational hours, the `log_in_attempts` table was filtered by time and status. The `WHERE` clause applies a greater-than operator (`>`) to isolate events occurring past 18:00. This is paired with an `AND` operator to ensure the query strictly returns entries where the authentication

failed, represented by the boolean value `0`

```
MariaDB [organization]> SELECT * FROM log_in_attempts WHERE login_time > '18:00'
 AND success = false;
+----------+----------+------------+------------+---------+----------------+---
------+
| event_id | username | login_date | login_time | country | ip_address     | su
ccess |
+----------+----------+------------+------------+---------+----------------+---
------+
|        2 | apatel   | 2022-05-10 | 20:27:27   | CAN     | 192.168.205.12 |
    0 |
|       18 | pwashing | 2022-05-11 | 19:28:50   | US      | 192.168.66.142 |
    0 |
|       20 | tshah    | 2022-05-12 | 18:56:36   | MEXICO  | 192.168.109.50 |
    0 |
|       28 | aestrada | 2022-05-09 | 19:28:12   | MEXICO  | 192.168.27.57  |
    0 |
|       34 | drosas   | 2022-05-11 | 21:02:04   | US      | 192.168.45.93  |
    0 |
|       42 | cgriffin | 2022-05-09 | 23:04:05   | US      | 192.168.4.157  |
    0 |
|       52 | cjackson | 2022-05-10 | 22:07:07   | CAN     | 192.168.58.57  |
    0 |
|       69 | wjaffrey | 2022-05-11 | 19:55:15   | USA     | 192.168.100.17 |
    0 |
|       82 | abernard | 2022-05-12 | 23:38:46   | MEX     | 192.168.234.49 |
    0 |
|       87 | apatel   | 2022-05-08 | 22:38:31   | CANADA  | 192.168.132.153 |
    0 |
|       96 | ivelasco | 2022-05-09 | 22:36:36   | CAN     | 192.168.84.194 |
    0 |
|      104 | asundara | 2022-05-11 | 18:38:07   | US      | 192.168.96.200 |
```

Retrieve login attempts on specific dates

Following a confirmed security anomaly on May 9, 2022, the search window was expanded to include the preceding 24 hours to check for preparatory reconnaissance traffic. Because SQL treats `OR` operations as distinct logical expressions, the query explicitly references the `login_date` column twice. This structural approach ensures the engine evaluates and pulls

records matching either targeted date without dropping rows

```
MariaDB [organization]> SELECT *
    ->
    -> FROM log_in_attempts
    ->
    -> WHERE login_date = '2022-05-09' OR login_date ='2022-05-08';
+----------+----------+------------+------------+---------+-----------------+-
------+
| event_id | username | login_date | login_time | country | ip_address      |
ccess |
+----------+----------+------------+------------+---------+-----------------+-
------+
|        1 | jrafael  | 2022-05-09 | 04:56:27   | CAN     | 192.168.243.140 |
    1 |
|        3 | dkot     | 2022-05-09 | 06:47:41   | USA     | 192.168.151.162 |
    1 |
|        4 | dkot     | 2022-05-08 | 02:00:39   | USA     | 192.168.178.71  |
    0 |
|        8 | bisles   | 2022-05-08 | 01:30:17   | US      | 192.168.119.173 |
    0 |
|       12 | dkot     | 2022-05-08 | 09:11:34   | USA     | 192.168.100.158 |
    1 |
|       15 | lyamamot | 2022-05-09 | 17:17:26   | USA     | 192.168.183.51  |
    0 |
|       24 | arusso   | 2022-05-09 | 06:49:39   | MEXICO  | 192.168.171.192 |
    1 |
|       25 | sbaelish | 2022-05-09 | 07:04:02   | US      | 192.168.33.137  |
    1 |
|       26 | apatel   | 2022-05-08 | 17:27:00   | CANADA  | 192.168.123.105 |
    1 |
|       28 | aestrada | 2022-05-09 | 19:28:12   | MEXICO  | 192.168.27.57   |
    0 |
```

Retrieve login attempts outside of Mexico

The threat intelligence team confirmed that the malicious activity did not originate from the organization's Mexican offices. However, due to inconsistent data formatting, Mexico is recorded as both "MEX" and "MEXICO" within the dataset. To account for this discrepancy, the `LIKE` operator was deployed alongside the `%` wildcard asset (`'MEX%'`). Prefacing this expression with a `NOT` operator systematically excludes all variations of

the country code, isolating international traffic for further review.

```
MariaDB [organization]> SELECT *
    ->
    -> FROM log_in_attempts
    ->
    -> WHERE NOT country LIKE 'MEX%';
+----------+----------+------------+------------+---------+-----------------+
------+
| event_id | username | login_date | login_time | country | ip_address      |
ccess |
+----------+----------+------------+------------+---------+-----------------+
------+
|        1 | jrafael  | 2022-05-09 | 04:56:27   | CAN     | 192.168.243.140 |
    1 |
|        2 | apatel   | 2022-05-10 | 20:27:27   | CAN     | 192.168.205.12  |
    0 |
|        3 | dkot     | 2022-05-09 | 06:47:41   | USA     | 192.168.151.162 |
    1 |
|        4 | dkot     | 2022-05-08 | 02:00:39   | USA     | 192.168.178.71  |
    0 |
|        5 | jrafael  | 2022-05-11 | 03:05:59   | CANADA  | 192.168.86.232  |
    0 |
|        7 | eraab    | 2022-05-11 | 01:45:14   | CAN     | 192.168.170.243 |
    1 |
|        8 | bisles   | 2022-05-08 | 01:30:17   | US      | 192.168.119.173 |
    0 |
|       10 | jrafael  | 2022-05-12 | 09:33:19   | CANADA  | 192.168.228.221 |
    0 |
|       11 | sgilmore | 2022-05-11 | 10:16:29   | CANADA  | 192.168.140.81  |
    0 |
|       12 | dkot     | 2022-05-08 | 09:11:34   | USA     | 192.168.100.158 |
    1 |
|       13 | mrah     | 2022-05-11 | 09:29:34   | USA     | 192.168.246.135 |
```

Retrieve employees in Marketing

To coordinate physical device upgrades for a specific subset of workstations, employee records were cross-referenced by division and physical location. The `AND` operator enforces a strict dual-parameter filter. It targets personnel assigned to the 'Marketing' department while simultaneously using the `LIKE 'East%'` wildcard pattern to capture any office number

configurations located within the East building complex.

```
MariaDB [organization]> SELECT *
    -> FROM employees
    -> WHERE department = 'marketing' AND office LIKE 'east%';
+-------------+--------------+----------+------------+----------+
| employee_id | device_id    | username | department | office   |
+-------------+--------------+----------+------------+----------+
|        1000 | a320b137c219 | elarson  | Marketing  | East-170 |
|        1052 | a192b174c940 | jdarosa  | Marketing  | East-195 |
|        1075 | x573y883z772 | fbautist | Marketing  | East-267 |
|        1088 | k8651965m233 | rgosh    | Marketing  | East-157 |
|        1103 | NULL         | randerss | Marketing  | East-460 |
|        1156 | a184b775c707 | dellery  | Marketing  | East-417 |
|        1163 | h679i515j339 | cwilliam | Marketing  | East-216 |
+-------------+--------------+----------+------------+----------+
7 rows in set (0.053 sec)
```

Retrieve employees in Finance or Sales

A targeted security configuration change required identifying all personnel within both the Finance and Sales teams. To execute a broad, multi-department pull, an `OR` operator was used to link two explicit condition statements. This logic ensures that if an employee's department

matches either specified parameter, their record is compiled into the deployment index.

```
MariaDB [organization]> SELECT *
    -> FROM employees
    -> WHERE department = 'Finance' OR department = 'sales';
+-------------+--------------+----------+------------+-------------+
| employee_id | device_id    | username | department | office      |
+-------------+--------------+----------+------------+-------------+
|        1003 | d394e816f943 | sgilmore | Finance    | South-153   |
|        1007 | h174i497j413 | wjaffrey | Finance    | North-406   |
|        1008 | i858j583k571 | abernard | Finance    | South-170   |
|        1009 | NULL         | lrodriqu | Sales      | South-134   |
|        1010 | k2421212m542 | jlansky  | Finance    | South-109   |
|        1011 | l748m120n401 | drosas   | Sales      | South-292   |
|        1015 | p611q262r945 | jsoto    | Finance    | North-271   |
|        1017 | r550s824t230 | jclark   | Finance    | North-188   |
|        1018 | s310t540u653 | abellmas | Finance    | North-403   |
|        1022 | w237x430y567 | arusso   | Finance    | West-465    |
|        1024 | y976z753a267 | iuduike  | Sales      | South-215   |
|        1025 | z381a365b233 | jhill    | Sales      | North-115   |
|        1029 | d336e475f676 | ivelasco | Finance    | East-156    |
|        1035 | j236k3031245 | bisles   | Sales      | South-171   |
|        1039 | n253o917p623 | cjackson | Sales      | East-378    |
|        1041 | p929q222r778 | cgriffin | Sales      | North-208   |
|        1044 | s429t157u159 | tbarnes  | Finance    | West-415    |
|        1045 | t567u844v434 | pwashing | Finance    | East-115    |
|        1046 | u429v921w138 | daquino  | Finance    | West-280    |
|        1047 | v109w587x644 | cward    | Finance    | West-373    |
|        1048 | w167x592y375 | tmitchel | Finance    | South-288   |
|        1049 | NULL         | jreckley | Finance    | Central-295 |
|        1050 | y132z930a114 | csimmons | Finance    | North-468   |
|        1057 | f370g535h632 | mscott   | Sales      | South-270   |
|        1062 | k3671639m697 | redwards | Finance    | North-180   |
|        1063 | l686m140n569 | lpope    | Sales      | East-226    |
|        1066 | o678p794q957 | ttyrell  | Sales      | Central-444 |
|        1069 | NULL         | jpark    | Finance    | East-110    |
|        1071 | t244u829v723 | zdutchma | Sales      | West-348    |
|        1072 | u905v920w694 | esmith   | Sales      | East-421    |
|        1076 | y347z204a710 | fgarcia  | Finance    | Central-270 |
|        1078 | a667b270c984 | sharley  | Sales      | North-418   |
```

Retrieve all employees not in IT

Following a rolling patch deployment, a compliance report was needed to identify staff members who had not yet received a critical system update. Because the Information Technology department had already been patched, the `NOT` operator was used to invert the

evaluation of the department column. This isolated all outstanding corporate directories while filtering out the IT department completely

```
MariaDB [organization]> SELECT * FROM employees WHERE NOT department = 'Info
+-------------+-------------+----------+-----------------+-------------+
| employee_id | device_id   | username | department      | office      |
+-------------+-------------+----------+-----------------+-------------+
|        1000 | a320b137c219 | elarson  | Marketing       | East-170    |
|        1001 | b239c825d303 | bmoreno  | Marketing       | Central-276 |
|        1002 | c116d593e558 | tshah    | Human Resources | North-434   |
|        1003 | d394e816f943 | sgilmore | Finance         | South-153   |
|        1004 | e218f877g788 | eraab    | Human Resources | South-127   |
|        1005 | f551g340h864 | gesparza | Human Resources | South-366   |
|        1007 | h174i497j413 | wjaffrey | Finance         | North-406   |
|        1008 | i858j583k571 | abernard | Finance         | South-170   |
|        1009 | NULL         | lrodriqu | Sales           | South-134   |
|        1010 | k2421212m542 | jlansky  | Finance         | South-109   |
|        1011 | l748m120n401 | drosas   | Sales           | South-292   |
|        1015 | p611q262r945 | jsoto    | Finance         | North-271   |
|        1016 | q793r736s288 | sbaelish | Human Resources | North-229   |
|        1017 | r550s824t230 | jclark   | Finance         | North-188   |
|        1018 | s310t540u653 | abellmas | Finance         | North-403   |
|        1020 | u899v381w363 | arutley  | Marketing       | South-351   |
|        1022 | w237x430y567 | arusso   | Finance         | West-465    |
|        1024 | y976z753a267 | iuduike  | Sales           | South-215   |
|        1025 | z381a365b233 | jhill    | Sales           | North-115   |
|        1026 | a998b568c863 | apatel   | Human Resources | West-320    |
|        1027 | b806c503d354 | mrah     | Marketing       | West-246    |
|        1028 | c603d749e374 | aestrada | Human Resources | West-121    |
|        1029 | d336e475f676 | ivelasco | Finance         | East-156    |
|        1030 | e391f189g913 | mabadi   | Marketing       | West-375    |
|        1031 | f419g188h578 | dkot     | Marketing       | West-408    |
|        1034 | i679j565k940 | bsand    | Human Resources | East-484    |
|        1035 | j236k3031245 | bisles   | Sales           | South-171   |
|        1036 | k5501533m205 | rjensen  | Marketing       | Central-239 |
|        1038 | m873n636o225 | btang    | Human Resources | Central-260 |
|        1039 | n253o917p623 | cjackson | Sales           | East-378    |
|        1040 | o783p832q294 | dtarly   | Human Resources | East-237    |
|        1041 | p929q222r778 | cgriffin | Sales           | North-208   |
|        1042 | q175r338s833 | acook    | Human Resources | West-381    |
|        1044 | s429t157u159 | tbarnes  | Finance         | West-415    |
|        1045 | t567u844v434 | pwashing | Finance         | East-115    |
```

Summary

Navigating large-scale security datasets requires structured, precise querying to minimize noise and surface actionable security data. By building and executing these targeted database queries, I demonstrated how to audit access logs, identify anomalous authentication timing, track geographic parameters despite inconsistent logging formats, and segment corporate

assets for risk mitigation. Mastering these core SQL filters directly enhances triage efficiency during security incidents, ensuring that critical network artifacts and system vulnerabilities are identified with minimal latency.